Recitation 1: Logic Gates, Number Systems, and 2’s Complement

CS-UY 2214

How Recitation Works

In order to get credit for recitation, you must

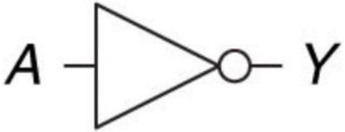
- Come to recitation (attendance is mandatory)
- Submit a something (make sure it's a plain text file)

How Rec Works:

- First 15-20 mins we will have a mini lesson to go over the material covered in the recitation
- The last 15-20 mins we will dedicate towards going over the answers to the recitation questions
- In order to leave early, you must get checked out by a TA and we will check for accuracy. Please ask to check out before the last 20 mins of recitation
- You must submit something by the end of recitation to get credit

Logic Gates

NOT



$$Y = \overline{A}$$

$Y = A'$

$Y = \neg A$

A	Y
0	1
1	0

AND

$$Y = AB$$

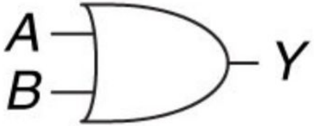
$Y = A \cdot B$

$Y = A \times B$

$Y = A \wedge B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR



$$Y = A + B$$

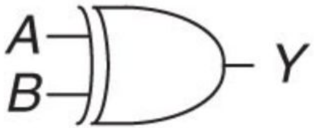
$Y = A \vee B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Note: The only thing that makes a NOT gate a NOT gate is the little circle, otherwise it is just A = Y

Logic Gates Cont.

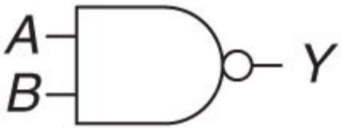
XOR



$$Y = A \oplus B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

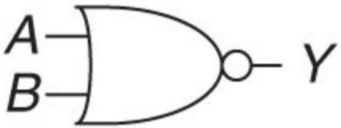
NAND



$$Y = \overline{AB}$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

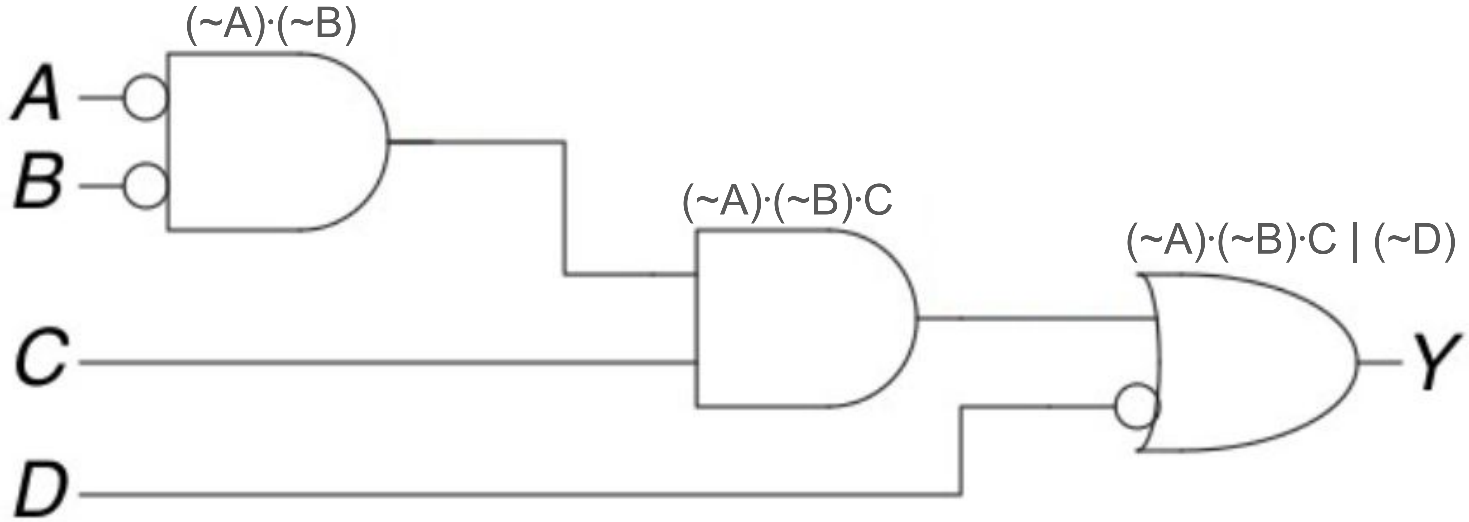
NOR



$$Y = \overline{A + B}$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Logic Gates Example



Y = (~A)·(~B)·C | (~D)

Number Systems

Base	**Name**	**Values**	**Prefix**
10	Decimal	0-9	0d (optional)
2	Binary	0&1	0b (necessary)
16	Hexadecimal	0-9, A-F	0x or 0h (necessary)

Decimal to Binary & Binary to Decimal

Convert 25 into Binary:

Division	Result	Remainder
25//2	12	1
12//2	6	0
6//2	3	0
3//2	1	1
1//2	0	1

Result: 0b11001 (arrange remainders from bottom to top)

Convert 0b11001 into Decimal:

1 1 0 0 1

2^4 2^3 2^2 2^1 2^0

16 + 8 +0 + 0 +1 = 25

Hexadecimal to Binary & Binary to Hexadecimal

Note: One Hexadecimal digit is equivalent to four Binary digits

Convert 0xA7F1 into Binary:

0xA = 10 = 0b1010

0x7 = 7 = 0b0111

0xF = 15 = 0b1111

0x1 = 1 = 0b0001

Result: 0b 1010 0111 1111 0001

Convert 0b 1101 1001 to Hexadecimal

0b1101 = 13 = 0xD

0b1001 = 9 = 0x9

Result: 0xD9

2’s Complement

Example of Preserving Band Width: Convert 20 into a 4 bit binary number.

20 = 0b**1**0100 = 0b0100 = 4

The band width has to be fixed in 2’s Complement to preserve addition and subtraction!

Representing Positive and Negative Numbers:

- If the MSB is 1, then the number is negative
- If the MSB is 0, then the number is positive

2’s Complement Steps & Example

Decimal to 2’s Complement:

Step 1. Convert the absolute value to binary

-3 → 3 → 0b0011

Step 2. If negative, invert the bits

0b1100

Step 3. If negative, add one to the inverted bits

1100 + 0001 = 0b1101

2’s Complement to Decimal:

Step 1. if the MSB is negative, invert all the bits

0b1101 → 0010

Step 2. Add one to the inverted bits

0010 + 0001 = 0b0011

Step 3. Convert result to decimal and add a minus sign

0b0011 = 3 → -3

Anubis Demo